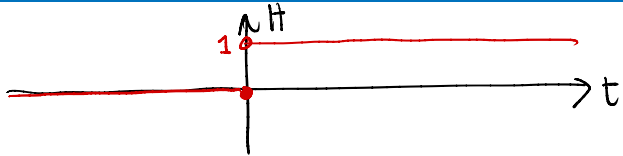
Lecture 12 recap



In Lecture 12:

- Learnt how the LT maps derivative in t-space to multiplication in s-space
- Understood how the LT encodes initial values
- Learnt how to solve IVPs using the LT and ILT
- Introduced the Heaviside step function
- Learnt how to use the s- and t-shifting theorems to calculate LTs and ILTs

$\mathcal{L}\{e^{\omega t} f(t)\} = F(s-\omega)$ || $\mathcal{L}\{f(t-c)H(t-c)\} = e^{-cs}F(s)$

Lecture 13

In Lecture 13, we will:

1. Learn how to solve IVPs involving a forcing that "switches on"
2. Introduce the Dirac delta

Discontinuous IVPs; **Exercise 3.17**: $y'(t) + y(t) = H(t-2)$, $y(0) = 1$

$H(t-2)$ on right-hand side $\Rightarrow$ forcing function switches on at $t = 2$.

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$$\underbrace{\mathscr{L}\{y'(t) + y(t)\}} = \mathscr{L}\{H(t-2)\}$$

$$\mathcal{L}\{y'\} + \mathcal{L}\{y\}$$

$$sY - \cancel{y(0)}^{1} \quad Y$$

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$$\underbrace{\mathscr{L}\{y'(t) + y(t)\}}_{sY - 1 + Y} = \mathscr{L}\{H(t-2)\}$$

$\mathcal{L}\{f(t-2)H(t-2)\} = e^{-2s}F(s) = e^{-2s}\frac{1}{s}$ $\quad f(t) = 1$

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$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} = \frac{A(s+1) + Bs}{s(s+1)} \Rightarrow \begin{array}{l} A + B = 0 \\ A = 1 \end{array}$$

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$$\Rightarrow \quad Y = \frac{1}{s+1} + \frac{\mathrm{e}^{-2s}}{s} - \frac{\mathrm{e}^{-2s}}{s+1}$$

$$y = \mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\} + \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\} - \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s+1}\right\} \qquad \text{linearity}$$

$$y = \underbrace{\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\}}_{\mathrm{e}^{-t}} + \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\} - \mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s+1}\right\} \qquad \text{linearity}$$

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$$\mathscr{L}^{-1}\{e^{-2s}F(s)\} = f(t-2)H(t-2) = e^{-(t-2)}H(t-2)$$

$$\mathscr{L}^{-1}\{F(s)\} = \mathscr{L}^{-1}\{1/(s+1)\} = e^{-t}$$

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$$= \begin{cases} e^{-t} & t \leq 2 \\ e^{-t} + 1 - e^{2-t} & t > 2 \end{cases}$$

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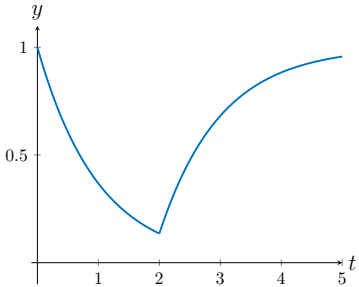

$$y' + y = H(t-2)$$

$$y' = -y + H(t-2)$$

$$= \begin{cases} -y & t \leq 2 \\ -y+1 & t > 2 \end{cases}$$

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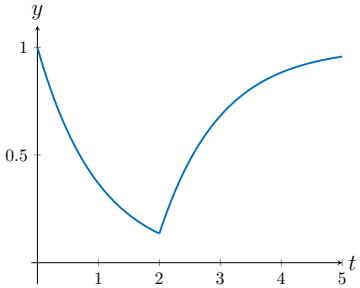
$$\lim_{t\to 2^-} y(t) = \lim_{t\to 2^+} y(t) = \mathrm{e}^{-2} \quad \text{but} \quad \lim_{t\to 2^-} y'(t) = -\mathrm{e}^{-2} \neq 1-\mathrm{e}^{-2} = \lim_{t\to 2^+} y'(t)$$



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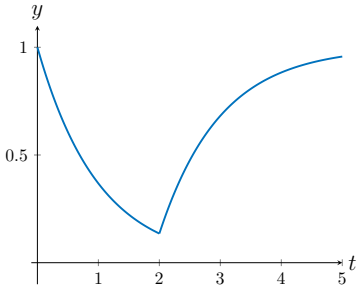

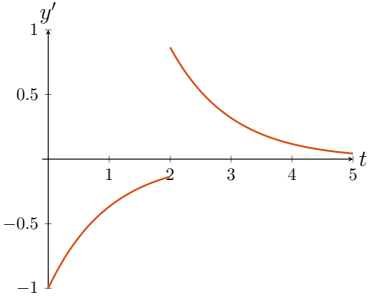

Solution to IVPs with discontinuous forcing

Consider the nth-order ODE

$$y^{(n)}(t) + f_{n-1}(t)y^{(n-1)} + \ldots + f_0(t)y(t) = g(t)$$

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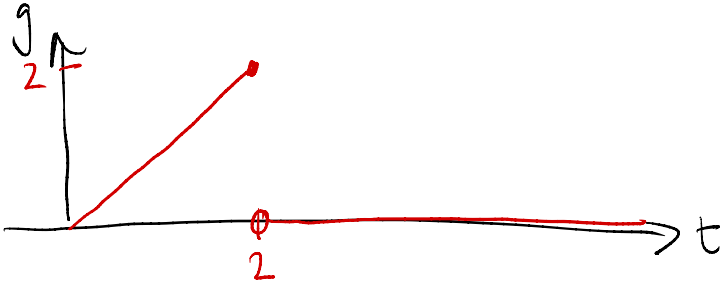
3. The left and right limits of $y^{(n)}$ are defined by the ODE at all discontinuities
4. y satisfies the ODE at all points of continuity of g

Exercise 3.18: Solve the below IVP

$$y'' + 4y = g(t) \quad \text{where} \quad g(t) = \begin{cases} t & 0 < t \leqslant 2 \\ 0 & t > 2 \end{cases} \quad \text{and} \quad y(0) = y'(0) = 0$$



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$$g(t) = t - t\,H(t-2) = t - 2\,\cancel{H(t-2)} - (t - \cancel{2})\,H(t-2)$$

Taking the Laplace transform of both sides:

$$\mathscr{L}\{y'' + 4\,y\} = s^2 Y + 4Y$$

$$\underbrace{\mathcal{L}\{y''\}}_{s^2 Y - \cancel{s y(0)} - y'(0)} + \underbrace{4\mathcal{L}\{y\}}_{4Y}$$

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$= \mathscr{L}\{f(t-2)H(t-2)\}$

where $f = t$

$\mathscr{L}\{t\} = 1/s^2$

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$$\Rightarrow \quad Y(s) = \frac{1}{s^2(s^2+4)} - \frac{2\,\mathrm{e}^{-2s}}{s\,(s^2+4)} - \frac{\mathrm{e}^{-2s}}{s^2\,(s^2+4)}$$

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Use partial fractions to write Y in a more convenient form

$$Y = \underbrace{\frac{1}{4s^2} - \frac{1}{4(s^2+4)}}_{= 1/s^2(s^2+4) \;=\; A/s \,+\, B/s^2 \,+\, \frac{C+Ds}{s^2+4}} - \mathrm{e}^{-2s}\left(\frac{1}{2s} + \frac{1}{4s^2} - \frac{s}{2(s^2+4)} - \frac{1}{4(s^2+4)}\right)$$

$$\Rightarrow (A+D)s^3 + (B+C)s^2 + 4A + 4B = 1$$

$$\Rightarrow B = +1/4 \Rightarrow C = -1/4$$

and $A = D = 0$

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$$Y = \frac{1}{4s^2} - \frac{1}{4(s^2+4)} - \mathrm{e}^{-2s}\left(\frac{1}{2s} + \frac{1}{4s^2} - \frac{s}{2(s^2+4)} - \frac{1}{4(s^2+4)}\right)$$

$$\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} = \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} - \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} - \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\}$$

$$- \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s^2}\right\} + \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{2\,s\,\mathrm{e}^{-2s}}{s^2+4}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2\,\mathrm{e}^{-2s}}{s^2+4}\right\}$$

Use partial fractions to write Y in a more convenient form

$$Y = \frac{1}{4s^2} - \frac{1}{4(s^2+4)} - \mathrm{e}^{-2s}\left(\frac{1}{2s} + \frac{1}{4s^2} - \frac{s}{2(s^2+4)} - \frac{1}{4(s^2+4)}\right)$$

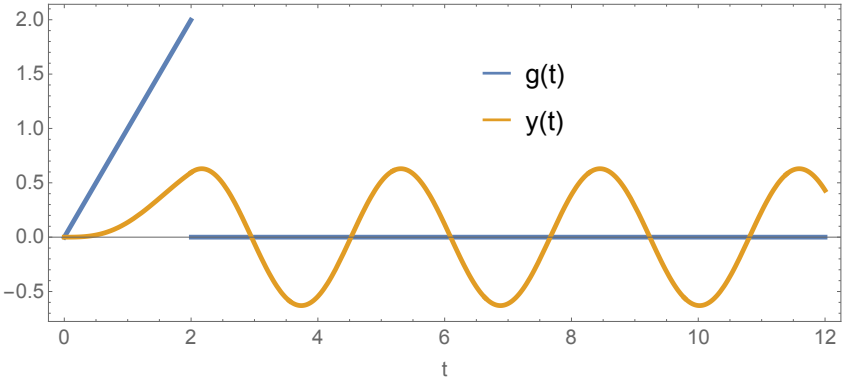
$$\Rightarrow \quad y(t) = \mathscr{L}^{-1}\{Y(s)\} = \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} - \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} - \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s}\right\}$$

$F(s) = \frac{1}{s}$

$$- \frac{1}{4}\mathscr{L}^{-1}\left\{\frac{\mathrm{e}^{-2s}}{s^2}\right\} + \frac{1}{\cancel{4}\,2}\mathscr{L}^{-1}\left\{\frac{\cancel{2}\,s\,\mathrm{e}^{-2s}}{s^2+4}\right\} + \frac{1}{8}\mathscr{L}^{-1}\left\{\frac{2\,\mathrm{e}^{-2s}}{s^2+4}\right\}$$

$$= \frac{1}{4}t - \frac{1}{8}\sin 2t$$

$$- \left(\frac{1}{2} + \frac{t-2}{4} - \frac{1}{\cancel{4}\,2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$

$$y(t) = \cancel{\frac{1}{4}t} - \frac{1}{8}\sin 2t - \left(\underbrace{\cancel{\frac{1}{2}} + \frac{t-\cancel{2}}{4}}_{\cancel{t/4}} - \frac{1}{\cancel{4}\,2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$

$$y(t) = \frac{1}{4}t - \frac{1}{8}\sin 2t - \left(\frac{1}{2} + \frac{t-2}{4} - \frac{1}{4}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$